AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318
Applied Network Security

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 22/Apr./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices questions- 1 mark per question. **Total: 10 Marks**

1. Which of the following is a symmetric encryption algorithm?
   a. RSA
   b. AES
   c. DSA
   d. Diffie-Hellman
2. The primary goal of encryption is to:
   a. Ensure data integrity
   b. Provide non-repudiation
   c. Maintain confidentiality
   d. Authenticate users
3. Which of the following attacks involves an attacker intercepting and altering communication?
   a. Eavesdropping
   b. Man-in-the-middle
   c. Replay attack
   d. Dictionary attack
4. In public key cryptography, the key used for encryption is:
   a. The same as the decryption key
   b. Kept private
   c. Publicly available
   d. Shared between users
5. The function of a hash algorithm is to:
   a. Encrypt data
   b. Decrypt data
   c. Produce a fixed-size output from variable-size input
   d. Sign data
6. Which of the following is an example of a block cipher mode of operation?
   a. ECB
   b. SHA
   c. RSA
   d. DES
7. Which cryptographic principle ensures that a message cannot be altered without detection?
   a. Confidentiality
   b. Integrity
   c. Authentication
   d. Non-repudiation
8. Which of the following is a passive attack?
   a. Denial of Service
   b. Traffic analysis
   c. Replay attack
   d. Masquerade
9. The Feistel cipher structure is used in DES.
   a. True
   b. False
10. The one-time pad is vulnerable to known-plaintext attacks.
    a. True
    b. False

Part 2: Answer all Three questions -short answer question **Total: 20 Marks**

Question (1) Answer the following questions (8 Marks): -

1. What is the difference between a block cipher and a stream cipher? (2 Marks)
2. Explain the concept of diffusion in cryptography. (2 Marks)

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3. Match each Feistel structure component with its function (4 Marks)

1. ECB (Electronic Codebook)	A) Encrypts counters; supports parallelism
2. CBC (Cipher Block Chaining)	B) Encrypts previous ciphertext block; propagates errors
3. CFB (Cipher Feedback)	C) No chaining, identical plaintext = identical ciphertext
4. CTR (Counter)	D) Converts block cipher into stream cipher with feedback

Question (2) Answer the following questions (8 Marks): -

1. Why is the one-time pad considered unconditionally secure? (2 Marks)
2. How does the meet-in-the-middle attack work against double DES? (2 Marks)
3. Match the following attacks with their descriptions (4 Marks):

1. Brute force	a. Exploits linear approximations of transformations
2. Differential cryptanalysis	b. Tries all possible keys
3. Linear cryptanalysis	c. Breaks double DES
4. Meet-in-the-middle	d. Observes how pairs of plaintext blocks evolve

Question (3) Answer the following questions (4 Marks): -

1. Explain the effectiveness of using layered security measures in protecting information systems. (2 Marks)
2. What is the purpose of the key expansion algorithm in AES? (2 Marks)

Part 3: Problem solving Questions **Total: 30 Marks**

1. Answer the following questions: (12 marks)
   a) Using Euclidian algorithm solve the following problems (4 Marks)
      1. 48 and 18 (2 Marks)
      2. 210 and 45. (2 Marks)
   b) Using Euler's Theorem and computing the Euler's totient function $\phi(n)$ for the following (4 Marks)
      1. ϕ (1800) (2 Marks)
      2. ϕ (323) (2 Marks)
   c) Find the multiplicative inverse using extended Euclidean for the following (4 Marks)
      1. 17 modulo 43.(2 Marks)
      2. 25 modulo 51. (2 Marks)
2. Answer the following questions on traditional cipher techniques: (9 marks)
   a) Given a plaintext "HELLO" and a key "KEY", encrypt it using the Caesar cipher with a shift of 3. (2 Marks)
   b) Decrypt the following ciphertext encrypted using the Vigenère Cipher with the keyword **KEY**: Ciphertext: **KXRKGIKXBKAL**(4 Marks)
   c) Decrypt the following ciphertext using the Rail Fence Cipher with depth 4: Ciphertext: HOEWRLOLLD(3 Marks)
3. Answer the following questions on Block cipher techniques: (9 marks)

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